

MA433 Homework 4

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Problem 5.2-2 on page 259 of Optimal Control

Minimum-time control of harmonic oscillator.

$$\ddot{y} + \omega_n^2 y = u(t)$$

It is desired to drive any initial states: $y(0) = x_{10}, \dot{y}(0) = x_{20}$ to zero in minimum time while the control is limited to: $|u(t)| \leq 1$ for all t .

Step I

Bolza Cost Form:

for $x_1 = y, x_2 = \dot{y}$:

$$\min J = \bar{\nu}^T \bar{\Psi} + \int_0^{t_f} 1 dt = \nu_1(x_1(t_f)) + \nu_2(x_2(t_f)) + \int_0^{t_f} 1 dt$$

subject to:

$$\begin{aligned} \text{continuous constraints: } \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u(t) - \omega_n^2 x_1 \end{aligned}$$

$$\begin{aligned} \text{terminal constraints: } x_1(t_f) &= 0 \\ x_2(t_f) &= 0 \end{aligned}$$

Step II

Hamiltonian:

$$H = L + \bar{\lambda}^T \bar{f} = 1 + [\lambda_{x_1} \quad \lambda_{x_2}] \begin{bmatrix} x_2 \\ u - \omega_n^2 x_1 \end{bmatrix} = 1 + \lambda_{x_1}(x_2) + \lambda_{x_2}(u - \omega_n^2 x_1)$$

Note: the Hamiltonian is linear in terms of $u(t)$, thus the optimal control must be either u lower or u upper.

Furthermore, a switching function is used in this case which is anything that multiplies $u(t)$ in H , thus:

$$S_f = \lambda_{x_2}$$

Euler-Lagrange Equations:

$$\dot{\bar{\lambda}}^T = -\frac{\partial H}{\partial \bar{x}} = -\begin{bmatrix} -\lambda_{x_2} \omega_n^2 & \lambda_{x_1} \end{bmatrix} = \begin{bmatrix} \lambda_{x_2} \omega_n^2 & -\lambda_{x_1} \end{bmatrix}$$

$$\bar{\lambda}_f^T(t_f) = \frac{\partial \Phi_f}{\partial \bar{x}_f} = \frac{\partial}{\partial \bar{x}_f} (\bar{\nu}_f^T \bar{\Psi}_f) = [\nu_1, \nu_2]$$

$\frac{\partial H}{\partial u}$ is omitted in favor of using the switching function.

Transversality Condition is required for free final time:

$$H(t_f) = \frac{\partial \Phi_f}{\partial t_f} = 0$$

Additionally, L and f are not explicitly functions of t , so $\dot{H} = 0$ making H constant for all time, thus $H(t) = 0$.

This means I can write:

$$H = 1 + \lambda_{x_1}(x_2) + \lambda_{x_2}(u - \omega_n^2 x_1) = 0$$

Because $u(t)$ is limited, Pontryagin's Minimum Principle states that H should be minimized over all $u(t)$.

Using the the above result from the transversality condition, to minimize H over all u , the following relationship

is noted:

$$\text{for } u > 1 \text{ then } S_f = \lambda_{x_2} < 1 \text{ and for } u < 1 \text{ then } S_f = \lambda_{x_2} > 1$$

This is to say that the switching function and the control signs are opposite to eachother for all time.

Step III

```

clc; clear; close all;
syms omega_n lambda_x_1(t) lambda_x_2(t) t_f nu_1 nu_2
assume(omega_n>0);
f = [diff(lambda_x_1,t)== lambda_x_2*omega_n^2; diff(lambda_x_2,t)== -lambda_x_1]

```

$$f(t) = \begin{pmatrix} \frac{\partial}{\partial t} \lambda_{x,1}(t) = \omega_n^2 \lambda_{x,2}(t) \\ \frac{\partial}{\partial t} \lambda_{x,2}(t) = -\lambda_{x,1}(t) \end{pmatrix}$$

```
cond = [lambda_x_1(t_f)==nu_1;lambda_x_2(t_f)==nu_2]
```

$$\text{cond} = \begin{pmatrix} \lambda_{x,1}(t_f) = \nu_1 \\ \lambda_{x,2}(t_f) = \nu_2 \end{pmatrix}$$

```

sol = dsolve([f,cond]);
lambda_x_1 = simplify(sol.lambda_x_1, 'steps', 100)

```

$$\text{lambda_x_1} = \nu_1 \cos(\omega_n (t - t_f)) + \nu_2 \omega_n \sin(\omega_n (t - t_f))$$

```
lambda_x_2 = simplify(sol.lambda_x_2, 'steps', 100)
```

$$\text{lambda_x_2} = \nu_2 \cos(\omega_n (t - t_f)) - \frac{\nu_1 \sin(\omega_n (t - t_f))}{\omega_n}$$

The switching function determines how many switching times are possible. The expression for the second Lagrange multiplier here shows that while theoretically infinite switching times are possible, the magnitude of the control means only one is necessary.

Step IV

Arc I

For the single switching time, this defines two arcs that must be integrated over in terms of τ_1, τ_2 .

$$\text{For Arc 1: } \tau_1 = \frac{t}{t_1}$$

$$\text{For Arc 2: } \tau_2 = \frac{t - t_1}{t_f - t_1}$$

```
syms x_1(t) x_2(t) t_1 x_10 x_20 u
assume(omega_n>0);
x_2 = diff(x_1,t)
```

$$x_2(t) = \frac{\partial}{\partial t} x_1(t)$$

$$f = \text{diff}(x_1, t, 2) + \omega_n^2 x_1 == -1$$

$$f(t) = \frac{\partial^2}{\partial t^2} x_1(t) + x_1(t) \omega_n^2 = -1$$

$$\text{cond} = [x_1(0) == x_{10}; x_2(0) == x_{20}]$$

$$\text{cond} = \left(\begin{array}{l} x_1(0) = x_{10} \\ \left(\left(\frac{\partial}{\partial t} x_1(t) \right) \Big|_{t=0} \right) = x_{20} \end{array} \right)$$

$$\text{sol} = \text{dsolve}(f, \text{cond})$$

$$\text{sol} = \frac{x_{20} \sin(\omega_n t)}{\omega_n} - \frac{1}{\omega_n^2} + \frac{\cos(\omega_n t) (x_{10} \omega_n^2 + 1)}{\omega_n^2}$$

$$x_{1_arc1} = \text{simplify}(\text{sol}, 'Steps', 100)$$

$$x_{1_arc1} = -\frac{2 \sin(0.5000 \omega_n t)^2 - \omega_n x_{20} \sin(\omega_n t)}{\omega_n^2} - x_{10} (2 \sin(0.5000 \omega_n t)^2 - 1)$$

$$x_{2_arc1} = \text{simplify}(\text{diff}(x_{1_arc1}, t), 'steps', 100)$$

$$x_{2_arc1} = x_{20} \cos(\omega_n t) - \frac{\sin(\omega_n t)}{\omega_n} - \omega_n x_{10} \sin(\omega_n t)$$

Step V

Arc II

```
syms x_1(t) x_2(t) t_f u_5
assume(omega_n>0);
x_2 = diff(x_1,t)
```

$$x_2(t) = \frac{\partial}{\partial t} x_1(t)$$

$$f = \text{diff}(x_1, t, 2) + \omega_n^2 x_1 == 1$$

$$f(t) = \frac{\partial^2}{\partial t^2} x_1(t) + x_1(t) \omega_n^2 = 1$$

$$\text{cond} = [x_1(t_f) == 0; x_2(t_f) == 0]$$

$$\text{cond} = \left(\begin{array}{l} x_1(t_f) = 0 \\ \frac{\partial}{\partial t_f} x_1(t_f) = 0 \end{array} \right)$$

$$\text{sol} = \text{dsolve}(f, \text{cond});$$

$$x_{1_arc2} = \text{simplify}(\text{sol}, 'Steps', 5)$$

$$x_{1_arc2} = -\frac{\cos(\omega_n (t - t_f)) - 1}{\omega_n^2}$$

$$x_{2_arc2} = \text{diff}(x_{1_arc2})$$

$$x_{2_arc2} = \frac{\sin(\omega_n (t - t_f))}{\omega_n}$$

Step VI

Let $\omega_n = 2 \text{ rad/s}$, $x_{10} = 0.25 \text{ m}$, $x_{20} = 0.25 \text{ m/s}$

```

x_1_arc1S = subs(x_1_arc1,[t,omega_n,x_10,x_20],[t_1,2,0.25,0.25])

x_1_arc1S =  $-\sin(t_1)^2 + 0.1250 \sin(2 t_1) + 0.2500$ 

x_2_arc1S = subs(x_2_arc1,[t,omega_n,x_10,x_20],[t_1,2,0.25,0.25])

x_2_arc1S =  $0.2500 \cos(2 t_1) - \sin(2 t_1)$ 

x_1_arc2S = subs(x_1_arc2,[t,omega_n,x_10,x_20],[t_1,2,0.25,0.25])

x_1_arc2S =  $0.2500 - 0.2500 \cos(2 t_1 - 2 t_f)$ 

x_2_arc2S = subs(x_2_arc2,[t,omega_n,x_10,x_20],[t_1,2,0.25,0.25])

x_2_arc2S =  $0.5000 \sin(2 t_1 - 2 t_f)$ 

times = vpasolve([x_1_arc1S==x_1_arc2S; x_2_arc1S==x_2_arc2S],[t_1 t_f],[0.5,1])

times = struct with fields:
    t_1: 0.3708
    t_f: 1.0619

t_1_val = double(times.t_1)

t_1_val = 0.3708

t_f_val = double(times.t_f)

t_f_val = 1.0619

```

Step VII

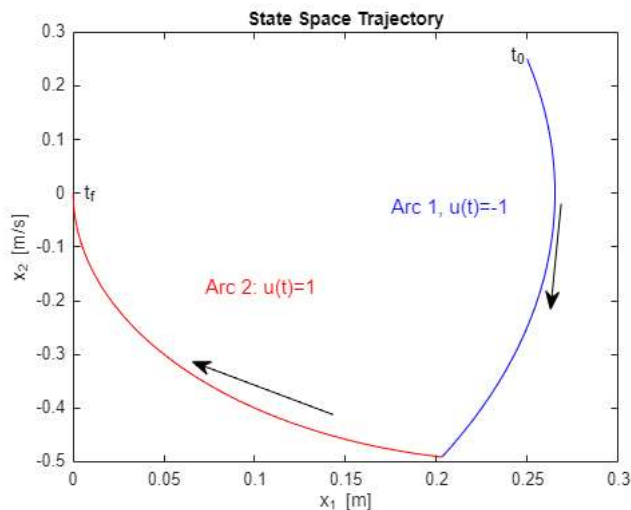
```

arc_1_time = linspace(0,t_1_val);
arc_2_time = linspace(t_1_val,t_f_val);
x_1_arc1val = subs(x_1_arc1,[omega_n, x_10, x_20],[2, .25, .25]);
x_1_arc1val = double(subs(x_1_arc1val,t,arc_1_time));
x_2_arc1val = subs(x_2_arc1,[omega_n, x_10, x_20],[2, .25, .25]);
x_2_arc1val = double(subs(x_2_arc1val,t,arc_1_time));

x_1_arc2val = subs(x_1_arc2,[omega_n, x_10, x_20, t_f],[2, .25, .25, t_f_val]);
x_1_arc2val = double(subs(x_1_arc2val,t,arc_2_time));
x_2_arc2val = subs(x_2_arc2,[omega_n, x_10, x_20,t_f],[2, .25, .25,t_f_val]);
x_2_arc2val = double(subs(x_2_arc2val,t,arc_2_time));

figure(1)
plot(x_1_arc1val,x_2_arc1val,'-b',x_1_arc2val,x_2_arc2val,'-r')
xlabel('x_1 [m]')
ylabel('x_2 [m/s]')
title('State Space Trajectory')
annotation('arrow',[0.825, 0.81],[0.6, 0.4]);
text(0.175, -0.025, 'Arc 1, u(t)=-1', 'FontSize', 12, 'Color', 'blue');
annotation('arrow',[0.5, 0.3],[0.2, 0.3]);
text(0.0725, -0.175, 'Arc 2: u(t)=1', 'FontSize', 12, 'Color', 'red');
text(0.24, 0.255, 't_0', 'FontSize', 12, 'Color', 'black');
text(0.005, 0, 't_f', 'FontSize', 12, 'Color', 'black');

```



Step VIII

Recall $H = 1 + \lambda_{x_1}(x_2) + \lambda_{x_2}(u - \omega_r^2 x_1) = 0$ thus at the final time $H(t_f) = 1 + \lambda_{x_2}(t_f)u(t_f) = 0$

From Step II also recall: $\bar{\lambda}_f^T(t_f) = \frac{\partial \Phi_f}{\partial x_f} = \frac{\partial}{\partial x_f}(\bar{\nu}_f^T \bar{\Psi}_f) = [\nu_1, \nu_2]$

Thus combining yields $1 + \nu_2 u(t_f) = 1 + \nu_2 = 0$ which gives $\nu_2 = -1$

Expression for the two Lagrange multipliers in terms of the additional adjoint multipliers are found in Step III.

```
lambda_x_1_t_1 = subs(lambda_x_1,[omega_n, t, t_f],[2 t_1_val,t_f_val])
```

```
lambda_x_1_t_1 = 0.1875 \nu_1 - 1.9645 \nu_2
```

```
lambda_x_2_t_1 = subs(lambda_x_2,[omega_n, t, t_f],[2 t_1_val,t_f_val])
```

```
lambda_x_2_t_1 = 0.4911 \nu_1 + 0.1875 \nu_2
```

Switching function crosses zero at switching time 1:

```
lambda_x_2_t_1 = 0==subs(lambda_x_2_t_1,nu_2,-1)
```

```
lambda_x_2_t_1 = 0 = 0.4911 \nu_1 - 0.1875
```

```
nu_1_val = double(solve(lambda_x_2_t_1,nu_1))
```

```
nu_1_val = 0.3818
```

```
nu_2_val = -1
```

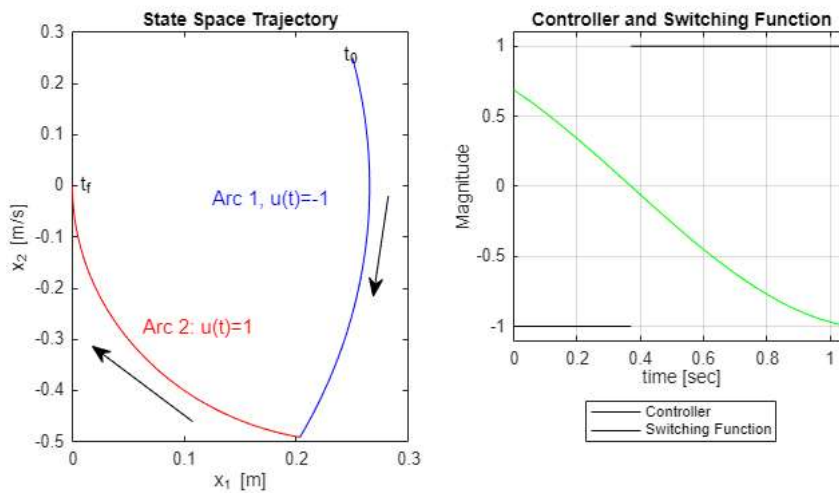
```
nu_2_val = -1
```

Step IX

```

figure(2)
grid on
subplot(1,2,1)
plot(x_1_arc1val,x_2_arc1val,'-b',x_1_arc2val,x_2_arc2val,'-r')
xlabel('x_1 [m]')
ylabel('x_2 [m/s]')
title('State Space Trajectory')
annotation('arrow', [0.445, 0.43], [0.6, 0.4]);
text(0.125, -0.025, 'Arc 1: u(t)=-1', 'FontSize', 12, 'Color', 'blue');
annotation('arrow', [0.25, 0.15], [0.15, 0.3]);
text(0.0625, -0.275, 'Arc 2: u(t)=1', 'FontSize', 12, 'Color', 'red');
text(0.24, 0.255, 't_0', 'FontSize', 12, 'Color', 'black');
text(0.005, 0, 't_f', 'FontSize', 12, 'Color', 'black');
subplot(1,2,2)
u_arc1 = -1*ones(length(arc_1_time),1);
u_arc2 = 1*ones(length(arc_1_time),1);
lambda_x_2val = subs(lambda_x_2,[nu_1,nu_2,t_f,omega_n],[nu_1_val,nu_2_val,t_f_val,2]);
time = linspace(0,t_f_val,200);
lambda_x_2val = double(subs(lambda_x_2val,t,time));
plot(arc_1_time,u_arc1,'-k',arc_2_time,u_arc2,'-k',time,lambda_x_2val,'-g')
ylim([-1.1 1.1])
grid on
set(gcf, 'Position', [100, 100, 800, 400]);
legend('Controller','Switching Function','Location','southoutside')
title('Controller and Switching Function')
xlabel('time [sec]')
ylabel('Magnitude')

```



Step X

$$J = \bar{\nu}^T \bar{\Psi} + \int_0^{t_f} 1 dt = \nu_1(x_1(t_f)) + \nu_2(x_2(t_f)) + \int_0^{t_f} 1 dt$$

```
J = int(1,t, 0, t_f_val)
```

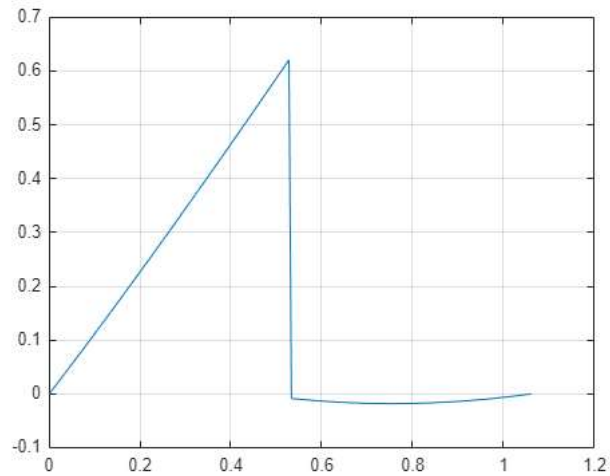
```
J = 1.0619
```

$$H = 1 + \lambda_{x_1}(x_2) + \lambda_{x_2}(u - \omega_r^2 x_1) = 0$$

```

lambda_x_1_val = subs(lambda_x_1,[nu_1 omega_n t_f nu_2],[nu_1_val 2 t_f_val nu_2_val]);
lambda_x_1_val = double(subs(lambda_x_1_val,t,time));
x_1_val = [x_1_arc1val, x_1_arc2val];
x_2_val = [x_2_arc1val, x_2_arc2val];
u_vals = [u_arc1; u_arc2]';
H = 1 + lambda_x_1_val.*(x_2_val)+lambda_x_2val.*(u_vals-4*x_1_val);
figure();
plot(time,H)
grid on

```



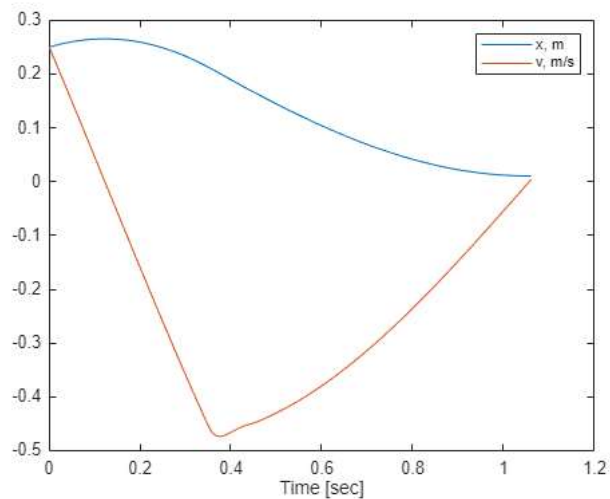
Step XI

```

x_10 = 0.25;
x_20 = 0.25;
t_1 = t_1_val;
t_f = t_f_val;

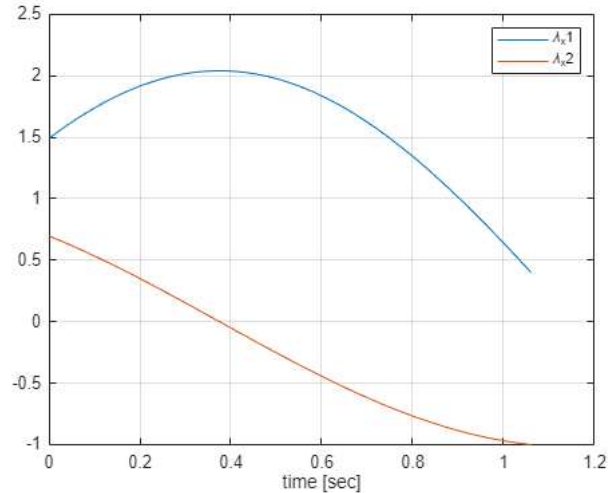
[T,S] = ode45(@(t,s)stateEOMs(t,s,t_1),linspace(0,t_f,201),[x_10, x_20]);
plot(T,S,'-')
xlabel('Time [sec]')
legend('x, m', 'v, m/s')

```



Lagrange Multiplier Guess

```
[T1,Lambda] = ode45(@LagrangeEOMs,T(end:-1:1),[0.4,-1]);
plot(T1,Lambda,'-')
xlabel('time [sec]')
legend('\lambda_x1', '\lambda_x2')
grid on
```



BVP4C

```
solinit.x = linspace(0,1,1000); %must go from zero to one because this is tau (1 and 2)
arc1 = interp1(T,[S,Lambda(end:-1:1,:)],t_1*solinit.x);
arc2 = interp1(T,[S,Lambda(end:-1:1,:)],t_1+(t_f-t_1)*solinit.x);
solinit.y = [arc1,arc2]'; %x_11 x_21 lx_11 lx_11 x_12 x_22 lx_12 lx_22 for all time, thus 8 rows by 21 columns (col determined by solinit.
solinit.parameters = [t_1,t_f];
tol = 1e-3;
sol = bvp4c(@ELEOMs,@(so,sf,ts)bcfunc(so,sf,ts,x_10,x_20,tol),solinit)
```

sol = struct with fields:

```
    solver: 'bvp4c'
         x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
         y: [8x12 double]
         yp: [8x12 double]
    parameters: [2x1 double]
         stats: [1x1 struct]
```

sol.parameters

```
ans = 2x1
    0.3698
    1.0625
```

Switching and final time agree with analytical solution.


```

for k = 4:11
    tol = 1*10^(-k);
    sol = bvp4c(@ELEOMs,@(so,sf,ts)bcfunc(so,sf,ts,x_10,x_20,tol),sol)
end

```

```

sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]
sol = struct with fields:
    solver: 'bvp4c'
    x: [0 6.1096e-08 1.2219e-07 8.0036e-06 1.5885e-05 3.1281e-05 0.1205 0.3106 0.5007 0.6905 0.8802 1]
    y: [8x12 double]
    yp: [8x12 double]
    parameters: [2x1 double]
    stats: [1x1 struct]

```

```
sol.parameters
```

```

ans = 2x1
    0.3708
    1.0619

```

```

figure()
grid on
subplot(1,2,1)
plot(sol.y(1,:),sol.y(2,:), '--b', sol.y(5,:), sol.y(6,:), '--r')
xlabel('x_1 [m]')
ylabel('x_2 [m/s]')
title('State Space Trajectory')

subplot(1,2,2)
u_vals = zeros(length(sol.x));

for k = 1:12
    if sol.x(k)<sol.parameters(1)
        u_vals(k) = -1;
    end
end

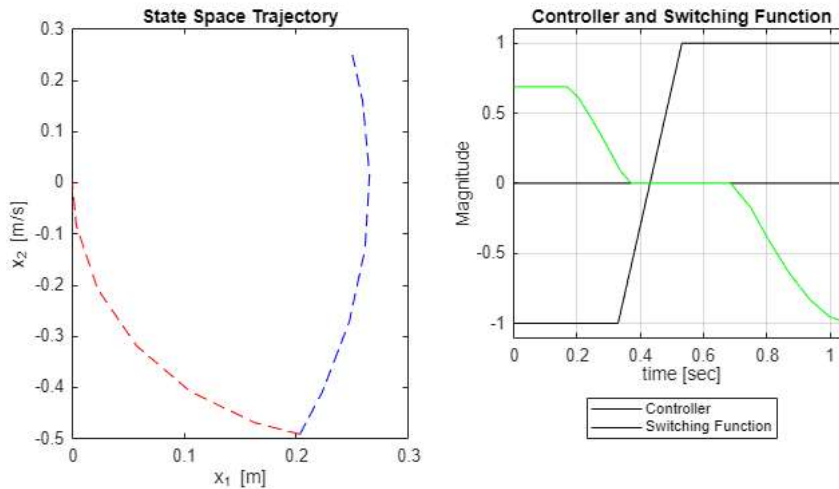
```

```

else
    u_vals(k) = 1;
end
end

plot(sol.x*sol.parameters(2),u_vals,'-k',linspace(0, sol.parameters(1),12),sol.y(4,:),...
'-g',linspace(sol.parameters(1), sol.parameters(2),12),sol.y(8,:),'-g')
ylim([-1.1 1.1])
grid on
set(gcf, 'Position', [100, 100, 800, 400]);
legend('Controller','Switching Function','Location','southoutside')
title('Controller and Switching Function')
xlabel('time [sec]')
ylabel('Magnitude')

```



There is good agreement between BVP4C and the analytical plots albiet the the last subplot for the numerica solution is funny despite starting and ending in the right places. I beleive this has to do with me making a mistake in the change of variable from time to tau and back.

Subfunctions

State EOMs used for guess generation

```

function dS = stateEOMs(t,s,t_1)
% s(1) = x_1 s(2) = x_2
if t < t_1
    u = -1;
else
    u = 1;
end
dS(1,1) = s(2);
dS(2,1) = u - (2^2)*s(1);
end

```

Lagrange EOMs used for guess generation

```

function dS = LagrangeEOMs(~,s)
% s(1) = l_x1 s(2) = l_x2
dS(1,1) = s(2)^2^2;
dS(2,1) = -s(1);
end

```

State EOMs with Optimal Control Law

```

function dS = StateEOMOptimal(s)
% s(1) = x_1 s(2) = x_2 s(3) = lambda_x1 s(4) = lambda_x2
if s(4) < 0
    u = 1;
else
    u = -1;
end
dS(1,1) = s(2);
dS(2,1) = u - (2^2)*s(1);
dS(3,1) = s(4)^2^2;
dS(4,1) = -s(3);
end

```

Euler-Lagrange EOMs used by BVP4C

```
function dS = ELEOMs(~,s,ts)
%ts =[t1,tf]
%arc 1
% s(1) = x_1 s(2) = x_2 s(3) = lambda_x1 s(4) = lambda_x2
%arc 2
% s(5) = x_1 s(6) = x_2 s(7) = lambda_x1 s(8) = lambda_x2
dS(1:4,1) = ts(1)*StateEOMOptimal(s(1:4));
dS(5:8,1) = (ts(2)-ts(1))*StateEOMOptimal(s(5:8));
end
```

Boundary Conditions Function

```
function con = bcfunc(so,sf,~,x_10,x_20,tol)
con(1) = so(1)-x_10;
con(2) = so(2)-x_20;
con(3:6) = sf(1:4)-so(5:8); %switching time states continuity
con(7) = sf(5)-tol;
con(8) = sf(6)-tol;
con(9) = sf(4); %lambda crosses zero at switching time
con(10) = 1+sf(7)*sf(6)+sf(8)*(1-4*sf(5));
end
```

$$H = 1 + \lambda_{x_1}(x_2) + \lambda_{x_2}(u - \omega_r^2 x_1) = 0$$